

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CE/IT) Examination

November 2013

IT401 Advanced Web Technology

Date: 12.11.2013, Tuesday

Time: 10 a.m. To 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q - 1

- (a) What are the features of using object oriented concepts in PHP? [03]
- (b) Write a script in PHP for retrieving data from XML and sending data to XML. [04]

Q - 2

- (a) Explain following Array functions in PHP with suitable example: [05]
 1. array_key_exists()
 2. array_slice()
 3. array_diff_assoc()
 4. array_chunk()
 5. array_shift()
- (b) Write function of MySQLi for following operation and explain it with example: [06]
 1. Fetches a result row as an associative, a numeric array, or both
 2. Returns the number of affected rows in the previous MySQL operation
 3. Adjusts the result pointer to an arbitrary row in the result-set

OR

- (b) 1. What is MySQLi extension in PHP? Explain the difference between mysql_connect() and mysqli_connect(). [06]
2. Write difference between mysqli_fetch_field_direct(), mysqli_fetch_field() and mysqli_fetch_fields().
- (c) What is Cookie? How can we set and destroy the cookie in PHP? Explain with small chunk of code in PHP. [03]

OR

- (c) What is Session? Explain use of Session for managing state in PHP. Take suitable example to explain your answer. [03]

Q - 3

- (a) 1. What is AJAX? How it works? Explain the complete life cycle of AJAX in detail. [05]
2. Explain XMLHttpRequest with example. [02]

OR

- (a) 1. Design complete login application in AJAX with validation. [05]
2. Display simple message "Hello" using AJAX when OnKeyUp event occur. [02]
- (b) 1. What is the difference between RSS feed and RSS aggregator? [04]
2. Write PHP script which uploads file with restrictions to the file extension. The user may upload .gif, .jpeg, and .png files; and the file size must be under 20kb. [03]

OR

- (b) 1. What is RSS? Explain advantages of RSS with example. [04]
2. Perform following tasks using PHP string functions [03]
i. Removes whitespace or other characters from the right end of a string.
ii. Splits a string into a series of smaller parts.
iii. Finds the last occurrence of a string inside another string

SECTION – II

Q - 4

- (a) Differentiate Syntactic Web and Semantic Web. How it is related to Web 2.0 and Web 3.0. [04]
(b) Define Web Service. How application can consume web service? Explain with small example. [03]

Q - 5

- (a) 1. Define following terms: [04]
a. Ontology
b. XML Namespace
c. XML Encoding
d. OWL
2. Define following functions: [03]
a. <xs:sort>
b. <xs:template>
c. <xs:sequence>
(b) Differentiate XML Schema with XML DTD. How both techniques perform XML Parsing for validating XML Document? Explain with example. [07]

OR

- (b) 1. What is XSLT? Write a code to define one XML file for Employee information, convert this XML file records in to HTML base table using XSLT. [05]
2. List various special characters for XML. How to define special characters in XML? [02]

Q - 6

- (a) 1. What is need of RDF? Compare RDF and XML. [03]
2. Give importance of RDFS? Justify following statement that "OWL is more expressive than RDFS". [04]
(b) 1. What is SOAP? Explain SOAP header part structure. [04]
2. Define Service. Compare service and program. [03]
OR
(b) 1. Prove or disprove "Restful web service preferred over SOAP based Web Service" [03]
2. What is need of WSDL file? Explain various element of it. [04]